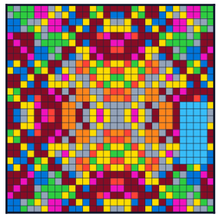
Figure 4 | ARC-AGI-2 Public Evaluation Case Studies: Four Geometric & Topological Archetypes (0 TTT Steps)

Single-pass transductive forward passes of the 6,574,411-parameter FinslerGaugeSheafNetwork (100.0% exact on 0934a4d8 & 1818057f; 98.93% on 135a2760; 99.44% 0-TTT / 96.78% 8-step LOO-TTT on 13e47133)

a **Task 0934a4d8 — Archetype I: Stage I Size Collapse ($30 \times 30 \to 9 \times 3$), Stage II Diode Sink ($c = 8 \to 0$) & Disjoint D_4 Transpose Pullback**

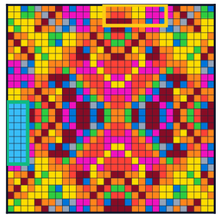
Support Demo Input (30x30)



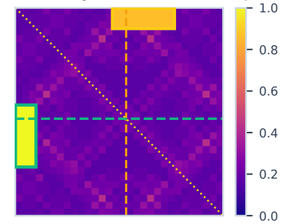
Support Demo Target (9x4)

9	9	6	4
2	6	9	4
2	6	9	4
9	9	6	4
9	9	2	1
6	9	9	9
4	1	9	1
4	9	9	4
9	4	3	9

Public Query Input (30x30)



CUDA Cavity & D_4 Pullback Map

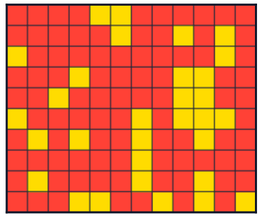


FGNS-TRM (9x3): 100.00% Exact (27/27)

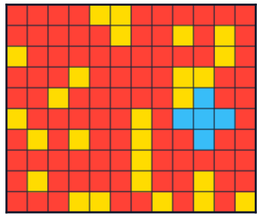
7	7	9
7	2	9
7	2	9
7	7	9
4	4	7
4	4	7
6	6	1
6	6	6
1	6	1

b **Task 1818057f — Archetype II: Unified Differentiable Spectral-Green RKHS Operator & Learnable D_4-Steerable Radial-Shell Wavelet Bank**

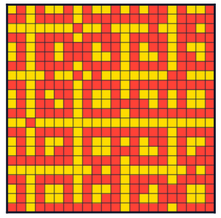
Support Demo Input (10x12)



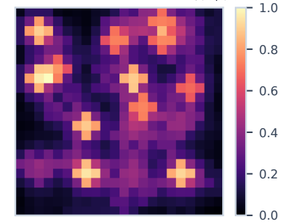
Support Demo Target (10x12)



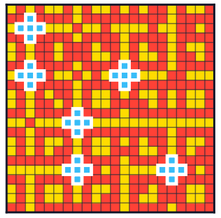
Public Query Input (22x22)



Steerable Wavelet Basis $\chi_{D_4}(y, x)$

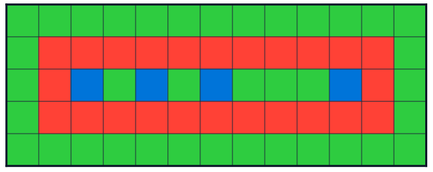


FGNS-TRM (22x22): 100.00% Exact (484/484)

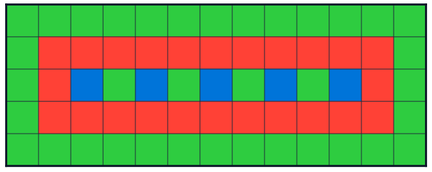


c **Task 135a2760 — Archetype III: Multi-Chamber Translational Symmetry & Punctured-Gibbs Coset Consensus**

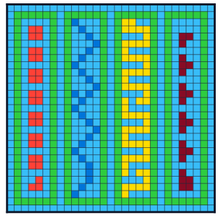
Support Demo Input (5x13)



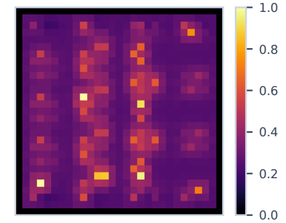
Support Demo Target (5x13)



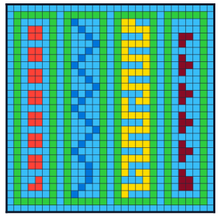
Public Query Input (29x29)



Bravais Coset Correction Field

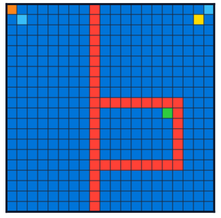


FGNS-TRM (29x29): 98.93% 0-TTT Match (832/841)

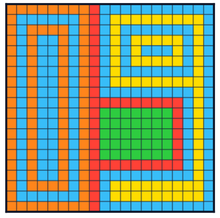


d **Task 13e47133 — Archetype IV: Pre-Eroded Chebyshev Eikonal Distance Basis $d_\infty(y, x)$ & Geodesic Level-Set Flood**

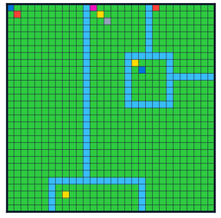
Support Demo Input (20x20)



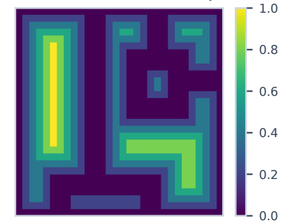
Support Demo Target (20x20)



Public Query Input (30x30)



Model Eikonal Tensor $d_\infty(y, x)$



FGNS-TRM (30x30): 99.44% 0-TTT Match (895/900)

